Modelling Particle Production in Analog Expanding Universes with High-Performance Computing

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Introduction

1. Big Bang and rapid expansion $\implies$ could observe particles
2. Big Bang was 14 billion years ago $\implies$ we can't observe particles
3. Condensed matter dynamics match particle dynamics (see Methods)
4. Can make condensed matter in the lab $\implies$ can effectively observe Big Bang particles
5. Lab based condensed matter system = "analog universe"

Related Work

Previous research has been done investigating a variety of different systems

1. “Hyper-phononic” quasiparticles and Lorentz Violation
2. Analog massive particles
3. Analog particles with non-zero quantum spin
4. Analog universes with different expansion behavior
5. Analog universes with non-zero background velocity
6. Experimental realizations (some extremely relevant and recent)

This Work

My focus is on

1. (3+1)D analog universe with high-performance computing
2. linear dispersion & phononic quasiparticle
3. analog massless particles
4. analog spin-0 particles
5. analog de Sitter expansion
6. no background velocity

Theory - Speed & Expansion

- Speed of sound in BEC = Speed of light in analog universe
- Speed decrease & static volume $\iff$ constant speed & expanding volume
- Example: $t = d/v$
  - Distance to grocery store doubles $\implies$ travel time doubles
  - Speed to the grocery store halves $\implies$ travel time doubles

Theory - The Field Equation

- The field equation describes the behavior of
  - a gas of quantum particles made in the lab
  - particles produced just after the Big Bang
- For each 3D wave vector **k** such that $||\mathbf{k}|| < k_c$ and $\hbar^2 k_c^2/2m = Un_0$
- Universe expansion starts and stops at finite times $\implies \eta_i \leq \eta \leq \eta_f$

The Field Equation

$$\partial_\eta^2 \widetilde{\theta}_\mathbf{k} - \frac{1}{2}\frac{\dot{a}(\eta)}{a(\eta)}\partial_\eta \widetilde{\theta}_\mathbf{k} - c_0^2 k^2 \widetilde{\theta}_\mathbf{k} = 0 \tag{1a}$$

$$\widetilde{\theta}_\mathbf{k}(0) = \widetilde{\theta}_{\mathbf{k}0} \quad \partial_\eta \widetilde{\theta}_\mathbf{k}(0) = -\frac{U_0}{\hbar} n_\mathbf{k} \tag{1b}$$

Theory - Numerical Integrator

- Conserve energy $\implies$ Use symplectic integrator
- a_m, b_m define optimality
- $\dfrac{\text{5th order McAte5 error}}{\text{4th order Ruth (1990) error}} \approx 3\%$

McAte5 Symplectic Integrator

$$1 \leq i \leq 5$$

$$\mathbf{p}_i = \mathbf{p}_0 + k \sum_{m=1}^{i} b_m \mathbf{F}(\mathbf{q}_{m-1}) \tag{2}$$

$$\mathbf{q}_i = \mathbf{q}_0 + k \sum_{m=1}^{i} a_m \mathbf{P}(\mathbf{p}_m) \tag{3}$$

Theory - High-Performance Computing

- Field equation for each $k < k_c$
- Solve in parallel with multiple GPUs
- Use the CWU "Turing" supercomputer
- 4x4 NVIDA P100 GPUs - Each 10.6 TFLOPS of single-precision (FP32) performance
- 180 GBps GPU-GPU bandwidth
- 80 GBps GPU-CPU bandwidth

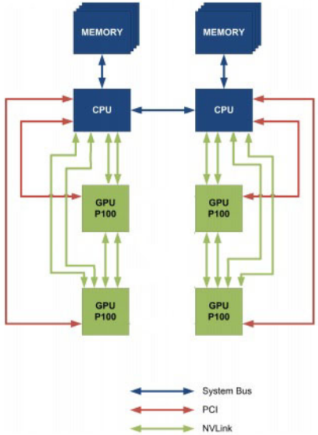

Figure: CPU to GPU and GPU to GPU interconnect using NVLink

Analytic Results - de Sitter Expansion

de Sitter expansion: $a(t) = e^{-t/t_s}$

- Population distribution highly depends on length of expansion
- Population is relatively high for low k
- Population drops off quickly with k

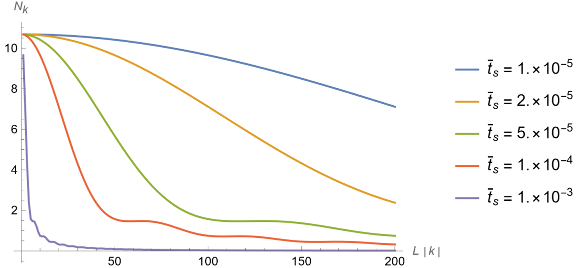

Figure: The momentum distribution when the universe stops expanding for multiple effective rates of expansion

Analytic Results - de Sitter Expansion

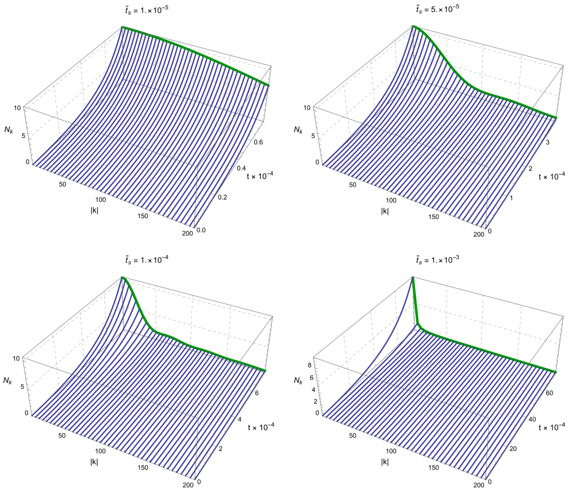

Figure: Momentum distribution throughout the expansion

Discussion - Why High-Performance Computing?

- Analytic methods $\implies$ simple expansions
- Numeric methods $\implies$ complex expansions
- Complex expansions $\implies$ High-performance computational model

Conclusion

- A lot of work has been done into the theory of analog universes for different systems
- A computational model is needed to provide guidance and support to both theory and experiment
- The behavior of particles created from the expansion of the universe is the same as particles in the lab
- Need to solve the equation of motion for many different parameters
- Need to support a general, complex expansion
- Use high-performance techniques and hardware for reasonable execution time

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Thank you & Code source

Thank you for your attention.

The code for this project is available at:

- GitHub (QR code): `github.com/nondairyneutrino/thesis`
- My website: `nathanwchapman.com`

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